

SARMASU, Romania

WMO: 151230

Lat: 46.7475N

Lon: 24.1597E

Elev: 1313

StdP: 14.01

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	13.2	3.4	6.9	10.5	8.4	8.9	15.3	20.6	37.3	17.2	32.5	2.2	320	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.6	88.6	68.6	85.8	67.8	82.8	66.6	71.1	84.2	69.6	82.2	68.2	80.0	5.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	103.5	76.6	65.5	98.8	74.5	64.2	94.5	72.8	35.7	84.7	34.4	82.4	33.2	80.1	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.4	12.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	27.7	34.1	41.7	51.2	60.7	66.9	68.5	70.1	60.9	50.9	41.7	31.5
	DBStd	16.20	8.04	8.31	7.75	9.02	6.22	5.68	5.21	5.89	8.00	7.02	8.43	7.33
	HDD50	2437	693	447	270	95	5	0	0	0	5	77	271	574
	HDD65	5765	1158	867	721	417	160	44	27	21	174	437	700	1039
	CDD50	2642	0	1	14	130	336	506	575	623	331	105	21	0
	CDD65	494	0	0	0	2	26	101	136	179	50	0	0	0
	CDH74	4260	0	0	0	49	230	836	1052	1675	413	5	0	0
	CDH80	1310	0	0	0	4	32	228	290	648	108	0	0	0
	WSAvg	4.5	3.7	4.7	5.5	6.3	4.8	4.2	4.0	3.9	4.4	3.9	4.1	4.1
	PrecAvg	22.1	0.9	0.8	1.1	1.8	2.5	3.4	3.2	1.9	2.1	1.7	1.3	1.2
(15) Precipitation	PrecMax	29.5	2.0	1.9	2.8	3.1	4.4	7.7	4.8	3.7	5.0	3.6	2.6	2.6
	PrecMin	15.2	0.3	0.2	0.2	0.1	0.7	0.6	0.9	0.3	0.3	0.0	0.0	0.2
	PrecStd	3.9	0.5	0.4	0.7	0.8	0.9	1.4	1.3	1.0	1.1	0.9	0.6	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	57.6	67.9	79.5	83.5	89.1	90.1	93.4	88.0	74.4	64.8	49.8
		MCWB	43.3	49.1	53.0	59.1	61.8	71.1	70.9	70.1	65.4	59.6	54.7	46.3
	2%	DB	42.9	52.6	62.2	74.2	79.7	85.9	86.3	89.8	82.8	71.0	60.3	46.0
		MCWB	40.4	46.1	49.1	57.6	62.6	69.1	69.4	67.8	64.3	58.2	52.6	43.3
	5%	DB	40.1	48.7	57.2	70.0	76.2	82.6	83.1	86.8	78.5	67.3	56.9	43.0
		MCWB	37.9	43.9	47.0	55.8	61.1	67.1	67.8	67.4	62.2	56.7	50.8	40.7
	10%	DB	37.3	45.3	53.0	66.0	72.7	79.9	79.9	83.4	74.1	62.9	53.7	40.0
		MCWB	35.5	41.5	44.4	53.4	59.7	65.8	66.6	66.9	61.3	54.6	48.7	38.4
	0.4%	WB	43.9	50.4	54.1	60.9	66.0	74.2	72.9	73.3	69.0	61.8	55.7	46.7
		MCDB	46.2	56.7	65.9	74.2	76.4	85.7	86.5	87.4	84.9	70.1	62.3	49.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.7	46.4	50.3	59.0	63.9	70.7	70.7	70.8	66.0	59.6	53.7	43.4
		MCDB	42.7	51.7	60.2	72.1	75.9	82.5	83.1	85.4	77.0	68.0	59.5	45.6
	5%	WB	38.1	44.0	47.8	57.2	62.3	68.8	69.3	69.2	64.1	57.6	51.7	40.9
		MCDB	39.8	48.0	55.7	68.1	73.7	80.2	80.5	83.3	74.8	66.1	56.5	42.7
	10%	WB	35.6	41.7	45.7	54.7	60.8	67.0	68.0	67.7	62.4	55.5	49.3	38.6
		MCDB	36.9	44.9	51.8	64.1	70.2	77.5	77.6	80.7	72.1	61.9	53.7	39.8
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	12.2	16.7	19.1	18.3	19.3	20.0	21.6	18.2	17.2	11.7	7.7
		MCDBR	10.4	16.9	23.5	25.7	23.5	24.4	24.2	26.7	24.4	24.9	16.6	11.3
		MCWBR	8.2	11.2	13.0	12.1	9.7	10.2	9.4	9.1	9.1	12.7	10.0	8.7
	5% WB	MCDBR	10.2	15.6	20.9	24.2	21.6	22.5	22.1	24.8	22.9	22.4	14.6	10.8
		MCWBR	8.3	10.8	12.2	11.6	9.6	10.0	9.3	9.5	9.2	11.9	9.1	8.7
(40) Clear-Sky Solar Irradiance	taub		0.309	0.335	0.375	0.420	0.405	0.402	0.421	0.425	0.387	0.359	0.336	0.311
	taud		2.336	2.300	2.255	2.166	2.268	2.305	2.255	2.243	2.319	2.384	2.396	2.367
	Ebn at Noon		245	261	267	264	273	274	266	260	260	250	233	230
	Edh at Noon		24	30	37	44	41	40	42	41	34	28	23	21
(44) All-Sky Solar Radiation	RadAvg		391	656	1013	1374	1685	1830	1807	1664	1183	805	454	300
	RadStd		54	78	131	138	160	150	157	152	137	100	61	39

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)	+1.35	N/A	N/A	N/A	+1.75	N/A	N/A	-283	-410	+166	+32

Nomenclature: See separate page